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Unit 6:

Cognition and Perception Pattern

Learning objectives

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- ❑ ***Up on completing this chapter you will be able to:***
 - ✓ Identify components of cognitive and thought processes
 - ✓ Describe normal X-tics of cognition
 - ✓ Recognize factors affecting normal cognitive function
 - ✓ Identify manifestations of altered thought process
 - ✓ use nursing process for clients experiencing altered thought process

NORMAL COGNITIVE FUNCTION

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- **Definition of terms**
- **Cognition:** systemic way in w/c person thinks, reasons, and uses language
- **Thought:** each instant of awareness (consciousness)
- **Memory:** capability of recalling thought at least once and usually again
- **Learning:** capability of nervous system to store memories
- The cerebral cortex coordinates consciousness, thoughts, memories, and learning which are inseparable functions

Normal Anatomic & Physiologic Function

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- Information perception precedes information processing
- Perception of information enters persons awareness through special senses and use four cognitive processes:-
 - ✓ *Consciousness,*
 - ✓ *Thoughts,*
 - ✓ *Memory, and*
 - ✓ *Learning*

Normal cognition...

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- Intact structures and perceptual functioning of cerebral cortex are needed for information processing
- Major sensory input is through eyes and ears.
- Visual information is carried to the visual cortex by neural axons from retinal ganglion cells w/c form optic nerve
- Sound is transmitted through auricle and ear canal to tympanic membrane then through ossicles w/c continue transmission to the inner ear
- Sound moves through fluid of inner ear to the apex of the cochlea and then to organ of cortex

Perception of information

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- Perception of reality includes: the awareness of time, the situation, and the self
- Perception is a complex phenomenon w/c depends on: functioning sense organs, neurotransmission, and intact central processing
- Senses contributing to perception: vision, hearing, touch, taste and smell

□ **Sensory receptors can be classified as:**

1. **Exteroceptors:** distance senses, include vision and hearing
2. **Proprioceptors:** near senses, include somatic senses (touch, temperature, and pain), taste and smell
3. **Interoceptors:** deep senses and arise from the viscera (internal organs) of the body and deeper tissues such as bone

Consciousness

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- A state of awareness and full responsiveness to stimuli
- Depends on intact RAS (reticular activating system) w/c divides in to facilitatory and inhibitory areas
- **Facilitatory area** is active and produce substances that increase muscle tone
- **Inhibitory area** causes opposite effect and hence sleep
- Consciousness is achieved well b/n the two areas are in equilibrium

Thoughts

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- Result of stimulation of many parts of the NS simultaneously

- The stimulation involves:
 - Cerebral cortex,
 - Thalamus,
 - Rhinoccephalon, and
 - The upper reticular formation of the brain

Classification of Memories

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- In connection with the prefrontal Lobes, another type of memory, called "**working memory**," w/c includes mainly short-term memory that is used during the course of intellectual reasoning but is terminated as each stage of the problem is resolved
- Memories are frequently classified according to the type of information that is stored
- One of these classifications divides memory into **declarative memory** and **skill memory**, as follows

- ❑ ***Declarative memory*** basically means memory of the various details of an integrated thought, such as memory of an important experience that includes
 - (1) Memory of the surroundings,
 - (2) Memory of time relationships,
 - (3) Memory of causes of the experience,
 - (4) Memory of the meaning of the experience, &
 - (5) Memory of one's deductions that were left in the person's mind

Memory...

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- **Skill memory** is frequently ass'ted with motor activities of the person's body, such as all the skills developed for hitting a tennis ball, including automatic memories to;
 - (1) Sight the ball,
 - (2) Calculate the r/ship & speed of the ball to the racquet, and
 - (3) Deduce (arrive at) rapidly the motions of the body, the arms, and the racquet (palm of the hands) required to hit the ball as desired—
- All of these activated instantly based on previous learning of the game of tennis—then moving on to the next stroke of the game while forgetting the details of the previous stroke

- (1) **Short-term memory:** w/c includes memories that last for seconds or at most minutes unless they are converted into longer-term memories
- (2) **Intermediate long-term memories:** w/c last for days to weeks but then fade away.
- (3) **Long-term memory:** w/c, once stored, can be recalled up to years or even a lifetime later

Memory

Two types of memory:

1. Short-term (temporary) memory:

- Retains data for only brief periods of time
- Information recorded is brief, shallow, and easily dissipated before it imprinted in to long-term memory.

2. Long-term (Lasting) memory

- Provides long term storage of cognitive data and holds up well with aging process
- Hippocampus, a brain part located in temporal lobes and part of limbic system, plays a great role in creating memories by preventing saved data from dissipating

Characteristics of Normal Cognition

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1. Intelligence

- Measurable product of intellectual functioning
- Consists *memory, comprehension, and concentration*
- **Comprehension** is capacity for understanding and reasoning
- **Concentration** is ability to screen out extraneous stimuli to focus on a task

Intelligence quotient (IQ)

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- Is the interpretation of mental age scores relatively to the chronological age
- The percentile of the MA & CA
- The ratio of mental age (MA) to chronological age (CA) multiplied by 100
- Ratio of mental age to chronological age growth

E.g. A child of CA-10 years & mental age 10 has an IQ of 100

- IQ – of 100 is considered as an average
- Used to compare people of different age

$$\text{IQ} = \frac{\text{mental age}}{\text{Chronological age}} \times 100\%$$

- conventional methods

$$\text{IQ} = \frac{\text{MA}}{\text{CA}} \times 100$$

- **Mental age (MA):** the performance of the average child at any chronological age
- Is the intelligence level of the child in terms of average chronological age of normal children with the same intellectual standards

- A child who passed all of the items at the five year level no other items had a mental age of five years.
- e.g. For 8 years child answers successfully all the test items placed at the age
- Is the measure of an individual level of intelligence at a given age
- **Chronological age (CA)** – The child's age in years or the numerical age of the children
- e.g. 8 years old, 10 years old, 18 years old

- ❑ ***The severity of mental retardation is ranging in to four level;***
- Mild mental retardation----- 50-70 (85%)
- Moderate mental retardation---- 35-50 (10%)
- Severe mental retardation----- 20-35 (4%)
- Profound mental retardation ----IQ < 20 (1%)

2. Reality perception (orientation)

- Awareness of time, place, situation and the self
- Knowledge of how the self and the environment interact in dimensions of place and time
- It depends on intactness of sense organs, neurotransmission, and central processing

3. Communication

- Use of language to store, process, and communicate (share) contents of thought.
- Language uses words w/c vary in use and meaning depending on person's age, education level, culture, socioeconomic background, and so on.

4. Judgment (Insight)

- Ability to process incoming stimuli and determine complex meaning that include many aspects of situation.
- Can be said process of reasoning.

5. Recall and Recognition

- Abilities used to retrieve information in the memories
- **Recall** is ability to retrieve data directly or by relating to other information
- **Recognition** is ability to accurately relate something currently in the env't with what is in the memory
- Both are skills that can be learned and need to be practiced to keep them actively useful

Normal Cognitive Functional Pattern

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I. Perceiving

- Process of recognizing and interpreting sensory stimuli that function as a basis for understanding, knowing or learning

II. Thinking

- Process of sorting, organizing, and categorizing information to form mental concepts or perception
- It can be said process of forming ideas or arriving at conclusions

Two types of thinking:

1. Concrete thinking

- Thinking that involves objects or group of things that can be perceived by senses.
- E.g. $1 + 1 = 2$ can be proofed by attaching the numbers to objects

2. Abstract thinking

- High level of thinking that involves a thought or idea that is apart from any material object.

III. Learning

- Process of acquiring knowledge.
- Multidimensional process that depends on symbols, language, classifications, concepts, and other concrete operations, along with abstract functioning.

IV. Remembering

- Process of retrieving data in memory for use.
- Memory is complex biochemical storage system that is not yet completely understood.
- Content of long-term memory depends on perceived value and significance of the event.
- Events like wedding and child birth hold greater value and memory potential.
- Short-term memory can easily affected by emotional stress.

Factors Affecting Normal Cognitive Function

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1. Adequate blood flow

- Adequate oxygenation depends on adequate blood flow.
- Oxygenation depends on three factors: respiratory function, circulatory function, and hemoglobin production
- Brain alone consumes 20% of the total oxygen uptake by the body and needs constant, ample supply to support brain cells' life.

2. Nutrition and metabolism

- Brain cells need constant supply of glucose for metabolic energy, iron for hemoglobin formation, and other nutrients for optimal functioning.
- Brain alone consumes 25% of glucose used by the body

3. Fluid and electrolytes balance

- Brain cells require constant extracellular env't of fluid and electrolytes for optimum functioning.
- Blood-brain barrier (BBB) protects brain cells from substances other than normal fluid and electrolytes that could cause damage

4. Sleep and rest

- Sleep has restorative function to regain energy for cognitive functions such as concentrating, organizing, remembering, and reasoning.
- REM (rapid eye mov't) sleep is particularly important for cognitive functioning.

5. Ability of organizing incoming stimuli

- Cognitive processes as perceiving, thinking, reasoning, and remembering depend on ability to receive and organize stimuli.
- Organizing function decline as age increase beyond adulthood

ALTERED COGNITIVE FUNCTION

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Potentials for altered cognition

- ❖ Inadequate blood flow
- ❖ Altered nutrition and metabolism
- ❖ Fluid and electrolytes imbalances
 - Can cause delirium (mental disorientation) or confusion (impairment of cognitive function)
- ❖ Infectious process
 - CNS infection, bacteremia, and septicemia can cause confusion
- ❖ Inadequate sleep and rest

❖ **Inability to organize incoming stimuli**

- Happen due to emotional stress or discomfort, env'tal stress, and pharmacologic agents.

❖ **Degenerative processes**

- Degeneration of brain cells ultimately affect cognitive processes
- Dementia: progressive brain deterioration x-zed by: confusion, disorientation, severely impaired; intellectual function, judgment, memory, planning, and problem-solving abilities.

Manifestations of altered cognitive function

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1. Disorganized thinking

- Disturbed thought process as manifested by regression, hallucinations, and delusions

2. Impaired thought processes

- Caused due to attention span deficits, memory impairments, level of arousal alterations, and sundown syndrome (disorientation and agitation that occur at night in institutionalized individuals)

3. Communication loss

- **Dysarthria** (difficulty in speaking, uncoordinated speech), caused by paralysis of the muscles responsible for producing speech
- Dysphasia or **aphasia** (defective speech or loss of speech), which can be **expressive aphasia**, **receptive aphasia**, or global (mixed) aphasia
- **Apraxia/dyspraxia** (inability to perform a previously learned action, inability to make skilled mov'ts with accuracy)

4. Perceptual disturbances

- Visual-perceptual dysfunctions, disturbances in visual-spatial relations, and sensory loss.
- **Hemianopsia** (loss of half of the visual field)
- Sensory loss
- Cognitive impairment and psychological effects

5. Motor loss

- **Hemiplegia** (paralysis of one side of the body)
- **Hemiparesis**, or weakness of one side of the body

ASSESSMENT

Subjective Data

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- **Functional pattern identification**
- Gather information on functional pattern of the client.
- Assess patients' usual cognitive function and its impact on activities of daily living.
- Assess perception of reality (orientation to place person and time).

- ❑ **Risk identification**
- ❑ Assess physiologic, psychological, and env'tal factors that can increase the likelihood of experiencing impaired thought processes.
- ❑ Assess existing and potential coping resources

☐ ***Physiologic risk factors:***

- Fluid imbalance
- Electrolytes imbalance
- Pain
- Decreased oxygen supply
- Renal or hepatic failure
- ✦ Renal or hepatic failure
- ✦ Inadequate blood flow
- ✦ Neurological impairment
- ✦ Systemic infection
- ✦ Medication toxicity

□ ***Psychological risk factors:***

- Grief response to losses
- Change in socioeconomic situation
- Depression or anxiety

□ ***Env'tal risk factors:***

- Unfamiliar env't
- Altered sensory input
- Mobility restriction

- ❑ **Dysfunction identification**
- ❑ Using data from functional pattern and risk identification
- ❑ Analyze and determine if dysfunction is present
- ❑ Provide clear information on identified dysfunctions

Objective data

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□ **Physical assessment**

Assess the following:

- Respiratory function
- Cardiovascular function
- Nutritional status
- Medical history
- Sleep and rest function
- Motor activity
- Activities of daily living

➤ **Diagnostic tests:**

- Physiologic tests

- Arterial oxygen
- Electrolytes
- Serum glucose
- Urea and toxic levels of drugs

-Tests of cognitive functions

NURSING DIAGNOSES

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1. **Altered thought process**

- Definition: disruption in cognitive operations and activities.
- Defining X-tics:
 - Inaccurate interpretation of env't
 - Disorganized thinking
 - Distractibility
 - Memory deficit/problems
 - Self-centeredness
 - Attention dysfunction
 - Non-reality based thinking

□ **Related factors**

- Physiologic factors
- Psychological/emotional factors
- Env'tal factors

□ **Related nursing diagnoses**

- Sensory/perceptual alterations
- Altered nutrition: less than body requirements
- Hypothermia
- Sleep pattern disturbances, etc

NURSING INTERVENTIONS

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- Nursing interventions to promote cognitive health and function
 - Promote healthy lifestyle
 - Memory improvement
 - Maintenance of learning skills

□ Nursing interventions for altered cognitive function

- Fluid intake and nutrition
- Mobility
- Safety
- Therapeutic communication
- Reality orientation
- Socialization therapy
- Recreation therapy
- Music therapy
- Family support

Normal perceptual Pattern

- ❑ Perception is a person's mental image or concept of elements in the environment including information gained through the senses.
- ❑ Cognition is emersion process of thinking and awareness system by which sensory inputs, past experiences and emotions are integrated and made meaningful.

Pain Perception

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- Pain is a highly unpleasant and very personal sensation that cannot be shared with others.
- It can occupy all a person's life.
- Pain is known to be the most common reason people seek help from health care workers.
- However no two people experience pain exactly the same way.

Pain Perception

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- In addition, the differences in individual pain perception and reaction, as well as the many causes of pain helps to develop plan to relieve pain and provide comfort.
- Effective pain management is an important aspect of care.

Physiologic factors that cause pain

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- Pain producing body substances (prostaglandins)
- Lack of endogenous opiates (Endorphins, enkephalin)
- Spinal reflexes injures
- Stimulation and reaction of pain receptors (Nociceptors)

Pain Perception

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- Normal patterns of body pain are the result of a complex pattern of stimuli generated at the pain site and transmitted to the brain.
- The perception is on sub cortical or cortical use of the brain.

Pain Perception

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- Sensory receptors of pain or nociceptors, are free nerve endings in the tissue that respond to physiologic stimuli like:
 - ▣ Temperature (thermoreceptors),
 - ▣ Chemical (chemo receptors),
 - ▣ position or movement changes (proprioceptors)
- Pain stimuli are transmitted subsequently from pain receptor from skin sensory nerves end to the thalamus and post central gyrus, where pain is perceived.

The Nature of Pain

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- Although pain is a universal experience, its exact nature remains a mystery.
- Different scholars defined pain in different ways but indicating that it is highly subjective and individual and that it is one of the body's defense mechanisms indicating that there is a problem.
- These concepts help caregivers to believe that the client is experiencing pain and that the client is the real authority on that pain.

Misconceptions & Bias related to pain

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- ❑ Clients experience severe pain only when they have had major surgery.
- ❑ The nurse or other health professionals are the authorities about a client's pain.
- ❑ Administering analgesics regularly for pain will lead to addiction.
- ❑ The amount of tissue damage is directly related to the amount of pain.
- ❑ Visible physiologic or behavioral signs accompany pain and can be used to verify its existence.

Origins and Causes of Pain

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- Pain can be categorized according to its origin as cutaneous, deep somatic, or visceral.
- Cutaneous pain originates in the skin or subcutaneous tissue.
- A paper cut causing a sharp pain with some burning is an example of cutaneous pain.
- Deep somatic pain arises from ligaments, tendons, bones, blood vessels, and nerves.
- It is diffuse and tends to last longer than cutaneous pain.

Origins and Causes of Pain

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- ❑ Visceral pain results from stimulation of pain receptors in the abdominal cavity, cranium and thorax.
- ❑ Visceral pain tends to appear diffuse and often feels like deep somatic pain that is, burning, aching, or a feeling of pressure.
- ❑ Stretching of the tissues, ischemia, or muscle spasms frequently causes visceral pain.

Type of pain

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- Based on its *duration* and intensity, it can be categorized as *acute* or *chronic*.
- **Acute pain** may have a sudden or slow onset; it varies from mild to severe, may last up to 6 months, and subsides as healing takes place.
- **Chronic pain** is prolonged, persistent non malignant pain that occurs over a 6 month period or longer.

Type of pain

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- Pain may also be described according to *where* it is experienced in the body.
- ***Radiating pain*** is perceived at the source of pain and extends to the nearby tissues.
- ***Referred pain*** is pain felt in a part of the body that is considerably removed from the tissues causing the pain.
- ***Intractable pain*** is pain that is highly resistant to relief. One example is the pain from an advanced malignancy.
- ***Phantom pain*** is a painful sensation perceived in a missing body part (e.g. an amputated leg) or in body part paralyzed from a spinal cord injury

Coping With Pain

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- Pain is now considered the fifth vital sign and is essential to regularly assess and manage in order to maintain the client's homeostasis and quality of life.
- The experience of pain is direct and personal.
- In this culture we tend to view pain as a negative condition and often go to any lengths to avoid the sensation.

Coping With Pain ...

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- The positive aspect of pain is that it is an early warning system; its presence triggers awareness that something is wrong in the body.
- Pain provides cues that allow us to modify our reactions and direct our behavior.
- One perspective of pain is that it is a message to our conscious self to check out any pain sensation before it gets worse.

Response to pain

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- The body's response to pain is a complex process rather than a specific action.
- It involves physiologic and psychosocial aspects.
- Initially the sympathetic nervous system responds, resulting the fight -or-flight response.
- As pain continues, the body adapts as parasympathetic nervous system takes over, reversing many of the initial physiologic responses.
- This adaptation occurs after several hours or days of pain.

Response to pain

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- The actual pain receptors adapt very little and continue to transmit the pain message.
- This serves the purpose of keeping the person continually aware the damaging stimuli causing the pain.
- The person may learn to cope with the pain through cognitive and behavioral activities, such as diversions, imagery, and excessive sleeping.
- The individual may respond to pain by seeking out physical interventions to manage the pain such as analgesics, massage and exercise.

Factors affecting the pain experience

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Ethnic/cultural Values

- In some cultures, pain may be considered a punishment for bad deeds; the individual therefore, to tolerate pain without complaint in order to atone for sins.
- Middle Eastern and African Cultures self-infliction of pain is a sign of mourning or grief.
- In other groups, pain may be anticipated as a part of the ritualistic practices of passage ceremonies, and therefore tolerance of pain signifies strength and endurance.

Factors affecting the pain experience

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Age

- Age is an important variable that influences how people admit or describe pain and how they behave.
- For example infants perceive pain and responds to pain with increased sensitivity.
- Toddler and preschooler often respond with crying and anger because child perceives pain as a threat to security.
- On the other side, adolescent may be slow to acknowledge pain and "giving in" may be considered weakness.

Factors affecting the pain experience

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Environment and support persons

- A strange environment such as a hospital, with its noises, lights and activity, can compound pain.
- Lonely person who is without a support network may perceive pain as severe, where as the person who has supporting people around may not perceive the pain as greatly.
- Some clients use pain to acquire secondary gains that is special attention from support persons and nurses.

Factors affecting the pain experience

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- Expectations of significant others can affect a person's perceptions of and responses to pain.
- In some situations, for example, girls may be permitted to express pain more openly than boys.
- Family role can also affect how a person perceives or responds to pain.
- For instance, a single mother supporting three children may ignore pain because of her need to stay on the job.
- The presence of support persons often changes a client's reaction to pain.

Factors affecting the pain experience

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Anxiety and stress

- Anxiety often accompanies pain.
- Threat of the unknown and the inability to control the pain or the events surrounding it often augment the pain perception.
- Fatigue also reduces a person's ability to cope, thereby increasing pain perception.
- When pain interferes with sleep, fatigue and muscle tension often result and increase the pain.

Normal Sensory Perception Pattern

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- A person's sensory perception depends on the sensory receptors and the reticular activating system as well as functioning nervous pathways to the brain.
- Stimuli are perceived through the five senses: sight, hearing, touch, smell and taste, although kinesthetic and visceral senses are stimulated internally.
- Sensory perception is a conscious process of selecting, organizing and interpreting sensory stimuli, and depends on intact function of sense organs, nervous pathway and brain.

Normal Sensory Perception Pattern

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Sensory awareness mechanisms

- ▣ Reticular activation system of brain's stem
- ▣ Intact nervous system and receptors
- ▣ Functional special sensory organ and somatic senses
- The characteristics of normal sensory perception are the normal measures in quality and quantity of the special and somatic senses.
- ▣ The normal vision of a healthy person is considered to be visual acuity at or near 20/20, full field vision and tricolor vision (red, green and blue).

Ctd...

E	1	20/200
F P	2	20/100
T O Z	3	20/70
L P E D	4	20/50
P E C F D	5	20/40
E D F C Z P	6	20/30
F E L O P Z D	7	20/25
D E F P O T E C	8	20/20
L E F O D P C T	9	
P D P L T C E O	10	
P E E L C F T D	11	

Normal pattern of body sensory perception

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- The normal hearing is auditory acuity of sounds at an intensity of 0 to 25 decibels, at frequencies of 125 to 8000 cycles per second.
- Taste is although discriminate sour, salty, sweet.
- The characteristic of somatic sense includes discriminating of touch, pressure, vibration, position, tickling, temperature and pain.

Factors affecting normal sensory perception

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- *Environment:* Can decrease or increase the sense of stimuli and response
- *Previous experience:* peoples are more alert to new stimuli that evoke a strong response that experience increase or decrease in response.

Factors affecting normal sensory perception

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- *Life styles and habits-* Life styles affect abundant stimuli, frequent changes, bright light, noise may prefer by someone and not by other for sense and response people with different life styles perceived stimuli differently.
- A functional disorder of the peripheral nervous system may result in sensory impairment (in case of chronic cigarette or alcohol).

Factors affecting normal sensory perception

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- *Illness* - Certain illness can affect sensory perception of individual like in case of diabetes hypertension as they cause changes in tiny blood vessels and nerves, leading to visual deficits and decreased sensation of touch in extremities.
- *Medications* – some antibiotics, including streptomycin and gentiamycin can cause damage to the auditory nerve, impairing hearing.

Altered Sensory Perception:

Potentials for altered sensors perception

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- Environnement and adaptation (noises, stranger)
- Internal factor (pain ,medication, anxiety, Worry)
- Age (dementia, old age)

Manifestation of altered sensory perception or deficit

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- *Anxiety*- irrational fear, isolation, fear
- *Cognitive dysfunction*-reasoning problem so diminished, reemerging loss
- *Hallucination*-sensory impression that are based on internal stimuli and have no basis in reality and distorted perception (senses)
- *Delusions*- beliefs that are not based on reality and reflect an unconscious need or fear (grandeur, persecutory)

Manifestation of altered sensory perception or deficit

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- Depression and withdrawal
- Hopelessness and loss of self-esteem
- Loss of contact with family, society and work
- Effective and perceptual disorders
- ***Presbyopia***- loss of accommodation of lens to focus light from near objects
- ***Cataract*** – an opacity of the lens and blocks light from reaching the retina
- ***Glaucoma***- increased in Intra ocular pressure, eye structures, including the optic nerve
- ***Presbycusis*** – Progressive loss of hearing acuity that occurs with advanced age leads sensoryneurial hearing loss

Manifestation of altered sensory perception or deficit

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- ***External otitis*** –infection of the skin in the external auditory canal
- ***Stroke (CVA)***- clot, hemorrhage or embolus obstructs blood flow in artery to the brain, causing Ischemia to tissue supplied by blood vessels
- ***Peripheral Neuropathy***- disorder of peripheral nerve lying outside spinal cord and brain damage to nerve can affect sensory and motor function.

Manifestation of altered sensory perception or deficit

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- Occupation and Leisure activities risk for sensory alteration
 - ▣ Hearing: factory, airport, construction worker.
 - ▣ Vision: welder, exposure to fire

Nursing Process for Pain Management: *Assessment*

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Pain History: this includes asking:

- ***Location***- where the pain occurred/ felt
- ***Intensity***- On a scale of 0 to 10 (with one representing the least pain level)
- ***Quality***- telling how one feels about pain
- ***Pattern***- the time of onset, duration, constancy
- ***Precipitating factors*** – factors that triggers or worsen the pain process.

Nursing Process for Pain Management: *Assessment*

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- ***Alleviating factors*** – factors lessening or relieving pain
- ***Associated symptoms*** – any symptoms occur with pain such as nausea, dizziness, blurred vision, shortness of breath before, during or after a pain
- ***Effects on activities of daily living*** – low pain affects daily of life such as eating, working, sleeping, social recreational activities.

Nursing Process for Pain Management: *Assessment*

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- *Past pain experience*
- *Coping resources* – what really done to cope with pain
- *Affective response* – how does the pain make one feel: anxious, depressed, frightened, tired, and burdensome.

Nursing Process for Pain Management: *Assessment*

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Physical examination

- To determine the client's physiologic and behavioral responses to pain, the nurse assesses the client's vital signs and observes the client for skin color changes, skin dryness, diaphoresis, facial expression and body gesture that reflect pain, discomfort, or anxiety, these findings provide valuable information about the severity of pain and how the client is coping with it.

Nursing Diagnosis

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- The nursing diagnosis given to clients experiencing pain is *pain* (implying acute pain) and chronic pain.
- When writing the diagnostic statement, the nurse may further specify the type of or location of the pain (e.g. postoperative, chest, abdominal, back).
- Because the pain experience affects so many facts of human functioning, pain itself may be the etiology of other nursing diagnoses.

Nursing Diagnosis

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- Examples of such nursing diagnoses are also shown below:
- Infective air way clearance related to postoperative incision chest
- Anxiety related to past experiences of poor control of pain and to anticipation of pain
- In effective breathing pattern related to postoperative abdominal pain

Nursing Diagnosis

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- Infective individual coping related to prolonged continuous back pain, ineffective pain management, and inadequate support systemic.
- Fear related to anticipated pain after surgery
- Altered health maintenance related to chronic pain and fatigue

Nursing Diagnosis

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- Hopelessness related to ineffective pain management strategies
- Knowledge deficit (pain control measures) related to lack of exposure to information resources
- Impaired physical mobility related to arthritic pain in knee and ankle joints
- Self-care deficit: Bathing /hygiene, dressing/ crooning, toileting related to pain in the joints
- Sleep pattern disturbance related to increased pain perception at night

Nursing Care Planning

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- The nurse identifies nursing interventions that will assist the client in achieving the overall client goals of preventing, modifying or eliminating pain so that the client is able partially or completely to resume usually daily activities and be able to cope more effectively with the pain experience.

Nursing Interventions

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- Pain management is the alleviation of pain or a reduction in pain to a level of comfort that is acceptable to the client.
- It includes two basic types of nursing interventions: *pharmacological* and *non pharmacological* interventions.
- Nursing management of pain consists of both independent and collaborative nursing actions.
- Generally speaking, a combination of strategies is best for the client in pain.

Nursing Interventions

Generally strategies for pain management

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- Acknowledge the client's pain – four ways of communicating this belief follow:
 1. Verbally acknowledge the presence of pain
 2. Listen attentively to what the client says about the pain
 3. Convey that you are assisting the client's pain to understand better, not to determine whether the pain is real
 4. Attend to the client's needs promptly
 5. Assisting support persons

Nursing Interventions

Generally strategies for pain management

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- Nurse can help by giving them accurate information about the pain and providing opportunities for them to discuss their emotional reactions, which may include anger, fear, frustration, and feelings of inadequacy.

Pharmacological pain Management

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- Pharmacologic pain management involves the use of opioids (narcotics), and non opioids (NSAIDs (nonsteroidal anti-inflammatory drugs), and adjuncts, or coanalgesic drugs.
- Non-opioids /NSAIDS include nonsteroidal anti-inflammatory drugs (NSAIDs) such as asprine, acetaminophen, and ibuprofen.
- These analgesics have anti-inflammatory, analgesic, and antipyretic effects.

Pharmacological pain Management

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- Adjuvant analgesics are medications that were developed for uses other than analgesia but have been found to reduce certain types of chronic pain in addition to their primary action.
- For example, mild sedatives or tranquilizers, such as diazepam may reduce painful muscle spasm as well as reduce anxiety, stress, and tension so that the client can obtain a good night sleep.

Non Pharmacologic pain Management

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- Non Pharmacologic pain management consists of a variety of **physical, cognitive or behavioral** pain management strategies.
- **Physical** intervention includes **cutaneous stimulation, immobilization, transcutaneous electrical nerve stimulation (TENS), acupuncture, and administration of placebos.**
- **Cognitive or behavioral** interventions include **destruction activities, relaxation techniques.**

Physical Interventions

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- The goals of physical intervention are:
 - ▣ Provide comfort
 - ▣ Correct physical dysfunction
 - ▣ Alter physiologic responses
 - ▣ Reduce fears associated with pain-related immobility or activity restriction

Cutaneous Stimulation

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- It provides temporary pain relief
- Distracts the client and focuses attention on the tactile stimulate away from the pain full sensations, thus reducing pain perception

Coetaneous Stimulation techniques include:

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- ▣ Massage
- ▣ Application of heat or cold
- ▣ Acupressure
- ▣ Contra lateral stimulation
- ▣ Cutaneous stimulation can be applied directly to the painful area, proximal to pain, distal to pain, and contra lateral (opposite side) to the pain.

Cognitive – Behavioral Interventions

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- The goals of cognitive – behavioral interventions are the following:
 - ▣ Alter pain perception
 - ▣ Alter pain behavior
 - ▣ Provide clients with greater sense of control over pain

Cognitive – Behavioral Interventions

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- *Distraction* – Distraction draws the person's attention away from the pain and lessens the perception of pain.
- In some instances, distraction can make a client completely unaware of pain.
- For example, while watching a football game on television, yet feel pain again the game is over.

Types of Distraction

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- ***Visual Distraction.*** Reading or watching TV, watching games, guided imagery
- ***Auditory Distraction.*** Humor (comedy, joking), listening to music,
- ***Tactile Distraction.*** Slow rhythmic breathing, massage, holding or stroking a pet or toy.
- ***Intellectual Distraction.*** Cross word puzzles, Card games, Hobbies (eg. writing a story).

Evaluation /Evaluating

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- Because pain is a subjective experience, most evaluative data collected are obtained by questioning the client about specific information taught; about level of comfort, level of energy, and sleep pattern; and whether modifications in activities or the use of non-invasive strategies were helpful.
- Some objective data may be obtained, particularly from the clients with acute pain (e.g. vital signs, behavioral postures and gestures).



!!!!THANK YOU